

CLUSTER ANALYSIS I

SESSION 8
COMP6140001 – DATA MINING

SUBJECT MATTER EXPERT
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LEARNING OBJECTIVE

L02 : apply various data mining techniques

L0 3: Apply data mining trends and research frontiers



OUTLINE

- Basic concepts
- K-Means
- K-Medoids

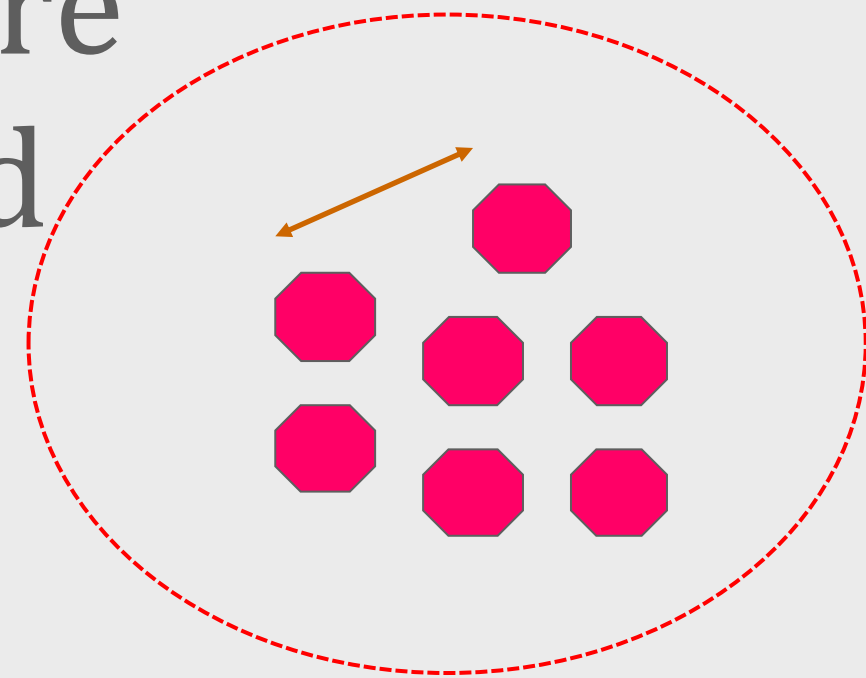
BASIC CONCEPTS



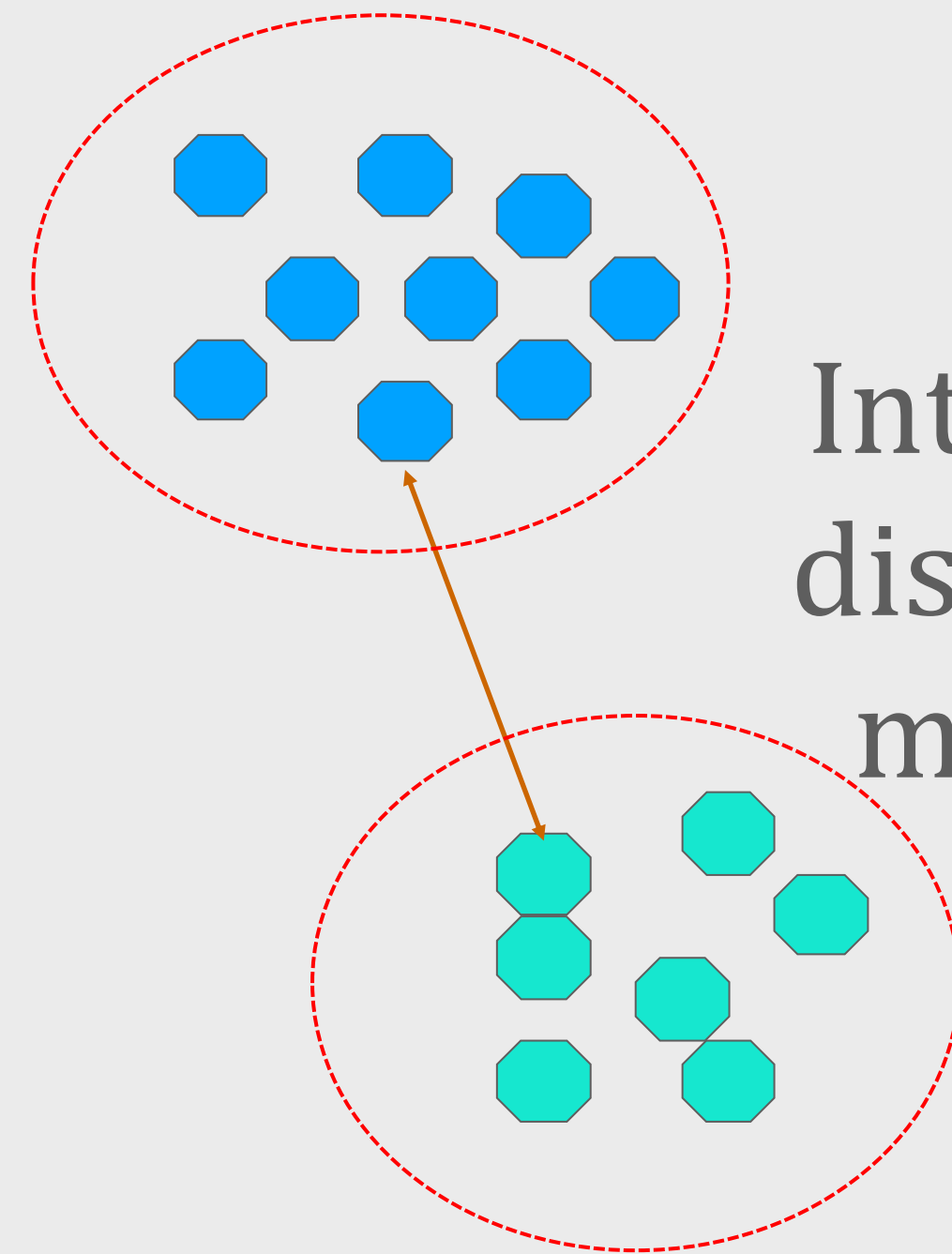
CLUSTER ANALYSIS: BASIC CONCEPTS

- **Cluster analysis** or **clustering** is the task of grouping a set of data objects in such a way that objects in the same group (called a cluster) are more similar (in some sense or another) to each other than to those in other groups (clusters).

Intra-cluster
distances are
minimized



Inter-cluster
distances are
maximized





CLUSTER ANALYSIS: BASIC CONCEPTS (2)

- Clustering can be achieved by various clustering algorithms that differ significantly in their notion of what constitutes a cluster and how to efficiently find them.
- Popular notions of clusters include groups with small distances among the cluster members, dense areas of the data space, intervals, or particular statistical distributions.
- Typical applications
 - As a stand-alone tool to get insight into data distribution
 - As a preprocessing step for other algorithms
- Cluster analysis is unsupervised learning, in which data objects are unlabeled, no predefined classes.



APPLICATIONS OF CLUSTER ANALYSIS

- Data reduction
 - Summarization: Reduce the size of large data sets, preprocessing for regression, PCA, classification, and association analysis
 - Compression: Image processing: vector quantization
- Hypothesis generation and testing
- Prediction based on groups
 - Cluster & find characteristics/patterns for each group
- Finding K-nearest neighbors
 - Localizing search to one or a small number of clusters
- Outlier detection: Outliers are often viewed as those “far away” from any cluster



WHAT IS NOT CLUSTER ANALYSIS?

- Supervised classification
 - Have class label information
- Simple segmentation
 - Dividing students into different registration groups alphabetically, by last name
- Results of a query
 - Groupings are a result of an external specification
- Graph partitioning
 - Some mutual relevance and synergy, but areas are not identical



QUALITY: WHAT IS GOOD CLUSTERING?

- A good clustering method will produce high quality clusters
 - high intra-class similarity: **cohesive** within clusters
 - low inter-class similarity: **distinctive** between clusters
- The quality of a clustering method depends on
 - the similarity measure used by the method
 - its implementation, and
 - its ability to discover some or all of the hidden patterns



TYPES OF CLUSTERING

- Hierarchical approach:
 - Create a hierarchical decomposition of the set of data (or objects) using some criterion
 - Typical methods: Diana, Agnes, BIRCH, CAMELEON
- Partitioning approach:
 - A division data objects into non-overlapping subsets (clusters) such that each data object is in exactly one subset.
 - Evaluate cluster by some criterion, e.g., minimizing the sum of square errors
 - Typical methods: k-means, k-medoids, CLARANS
- Density-based approach:
 - Based on connectivity and density functions
 - Typical methods: DBSACN, OPTICS, DenClue
- Grid-based approach:
 - based on a multiple-level granularity structure
 - Typical methods: STING, WaveCluster, CLIQUE

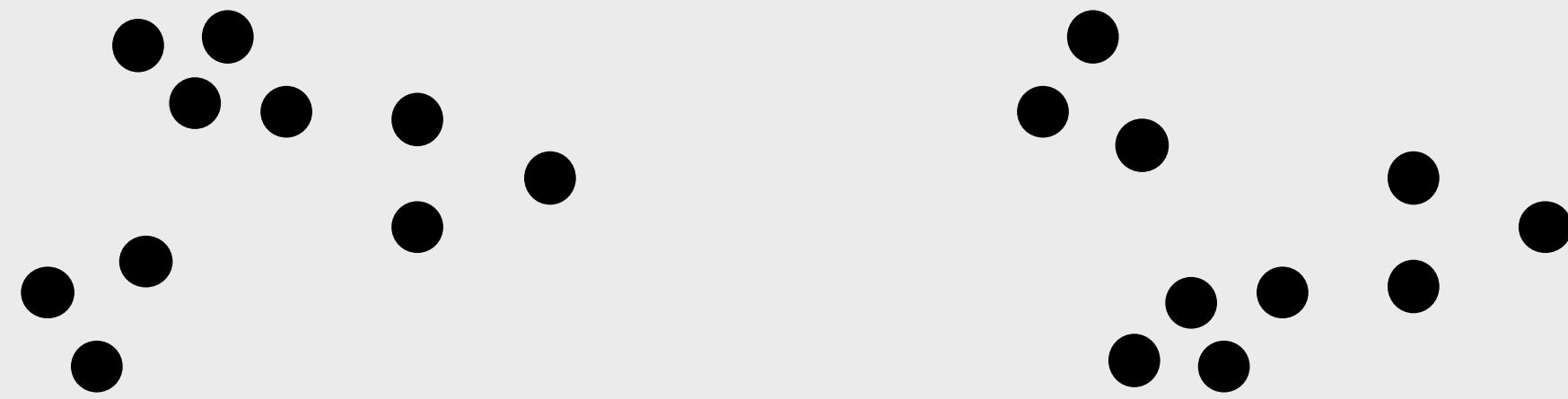


TYPES OF CLUSTERING (2)

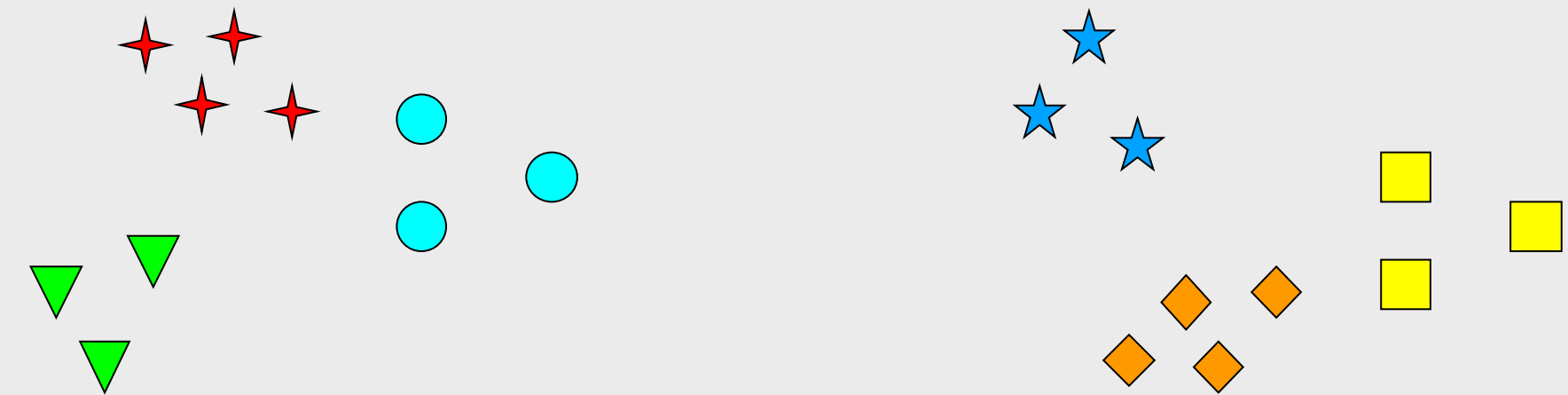
- **Model-based:** A model is hypothesized for each of the clusters and tries to find the best fit of that model to each other
 - Fuzzy Clustering
 - Mixture Model
 - Self-Organizing Map (SOM)
- **Frequent pattern-based:** Based on the analysis of frequent patterns
 - Typical methods: p-Cluster
- **User-guided or constraint-based:** Clustering by considering user-specified or application-specific constraints
 - Typical methods: COD (obstacles), constrained clustering
- **Link-based clustering:**
 - Objects are often linked together in various ways
 - Massive links can be used to cluster objects: SimRank, LinkClus



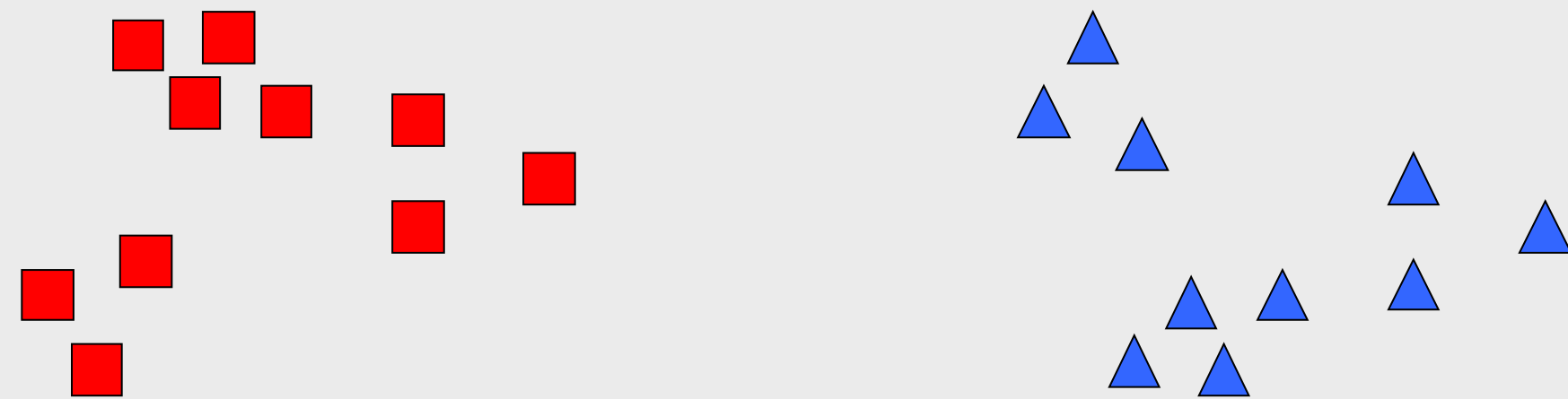
NOTION OF A CLUSTER CAN BE AMBIGUOUS



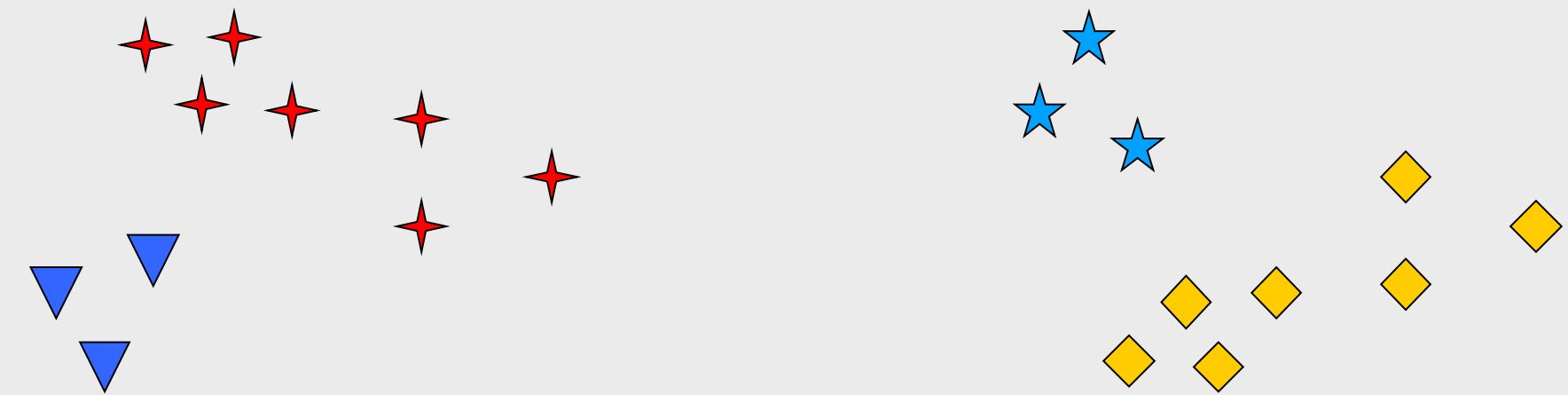
How many clusters?



Six Clusters



Two Clusters



Four Clusters



CHARACTERISTICS OF THE INPUT DATA ARE IMPORTANT

- Dimensionality
- Noise and Outliers
- Type of Distribution
- Type of proximity or density measure
 - This is a derived measure, but central to clustering
- Sparseness
 - Dictates type of similarity
 - Adds to efficiency
- Attribute type
 - Dictates type of similarity
- Type of Data
 - Dictates type of similarity
 - Other characteristics, e.g., autocorrelation



PROXIMITY MEASURE

- Refers to a data similarity and dissimilarity measure
- **Similarity**
 - Numerical measure of how alike two data objects are
 - Value is higher when objects are more alike
 - Often falls in the range $[0,1]$
- **Dissimilarity** (e.g., distance measure)
 - Numerical measure of how different two data objects are
 - Lower when objects are more alike
 - Minimum dissimilarity is often 0
 - Upper limit varies



PROXIMITY MEASURE (2)

- **Data matrix**
 - Used to store the data objects
 - n data points with p dimensions
 - Two modes
- **Dissimilarity matrix**
 - Used to store dissimilarity values for pairs of objects
 - n data points, but registers only the distance
 - A triangular matrix
 - Single mode
- Object dissimilarity **computation** based on the **attribute type** of objects

$$\begin{bmatrix} x_{11} & \dots & x_{1f} & \dots & x_{1p} \\ \dots & \dots & \dots & \dots & \dots \\ x_{i1} & \dots & x_{if} & \dots & x_{ip} \\ \dots & \dots & \dots & \dots & \dots \\ x_{n1} & \dots & x_{nf} & \dots & x_{np} \end{bmatrix}$$

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & & \\ d(n,1) & d(n,2) & \dots & \dots & 0 \end{bmatrix}$$



PROXIMITY MEASURE (3)

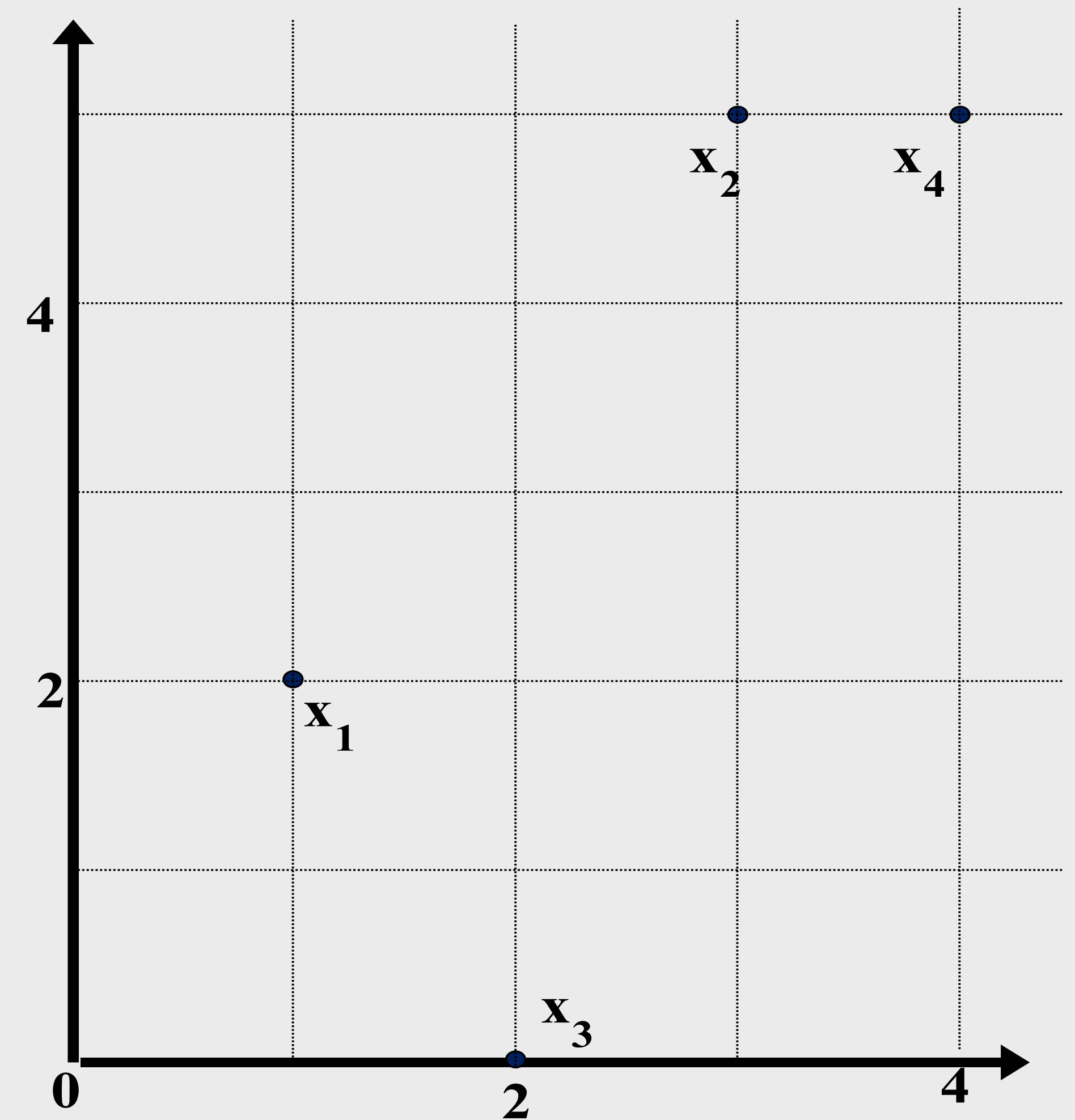
- Example:

- Data matrix

Point	Attr. 1	Attr. 2
x_1	1	2
x_2	3	5
x_3	2	0
x_4	4	5

- Dissimilarity matrix (with Euclidean distance)

	x_1	x_2	x_3	x_4
x_1	0			
x_2	3.61	0		
x_3	2.24	5.1	0	
x_4	4.24	1	5.39	0





PROXIMITY MEASURE FOR NUMERIC ATTRIBUTES

- **Minkowski distance:**

$$d(i, j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \cdots + |x_{ip} - x_{jp}|^h}$$

where $i = (x_{i1}, x_{i2}, \dots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jp})$ are two p -dimensional data objects, and h is the order (the distance so defined is also called $L - h$ norm)

- Properties

- $d(i, j) > 0$ if $i \neq j$, and $d(i, j) = 0$ (*positive definiteness*)
- $d(i, j) = d(j, i)$ (*symmetry*)
- $d(i, j) \leq d(i, k) + d(k, j)$ (*triangle inequality*)

- A distance that satisfies these properties is a **metric**



PROXIMITY MEASURE FOR NUMERIC ATTRIBUTES (2)

- Special case of **Minkowski** distance:
- $h = 1$, **Manhattan** (city block, L_1 norm) distance

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \cdots + |x_{ip} - x_{jp}|$$

- $h = 2$, **Euclidean** distance (L_2 norm)

$$d(i, j) = \sqrt{|x_{i1} - x_{j1}|^2 + |x_{i2} - x_{j2}|^2 + \cdots + |x_{ip} - x_{jp}|^2}$$

- $h = \infty$, **supremum** distance (L_{max} norm, L_∞ norm)

$$d(i, j) = \lim_{h \rightarrow \infty} \left(\sum_{f=1}^p |x_{if} - x_{jf}|^h \right)^{\frac{1}{h}} = \max_f |x_{if} - x_{jf}|$$

PROXIMITY MEASURE FOR NUMERIC ATTRIBUTES (3)

- Example:

Point	Attr. 1	Attr. 2
x_1	1	2
x_2	3	5
x_3	2	0
x_4	4	5

Data matrix

L_2	x_1	x_2	x_3	x_4
x_1	0			
x_2	3.61	0		
x_3	2.24	5.1	0	
x_4	4.24	1	5.39	0

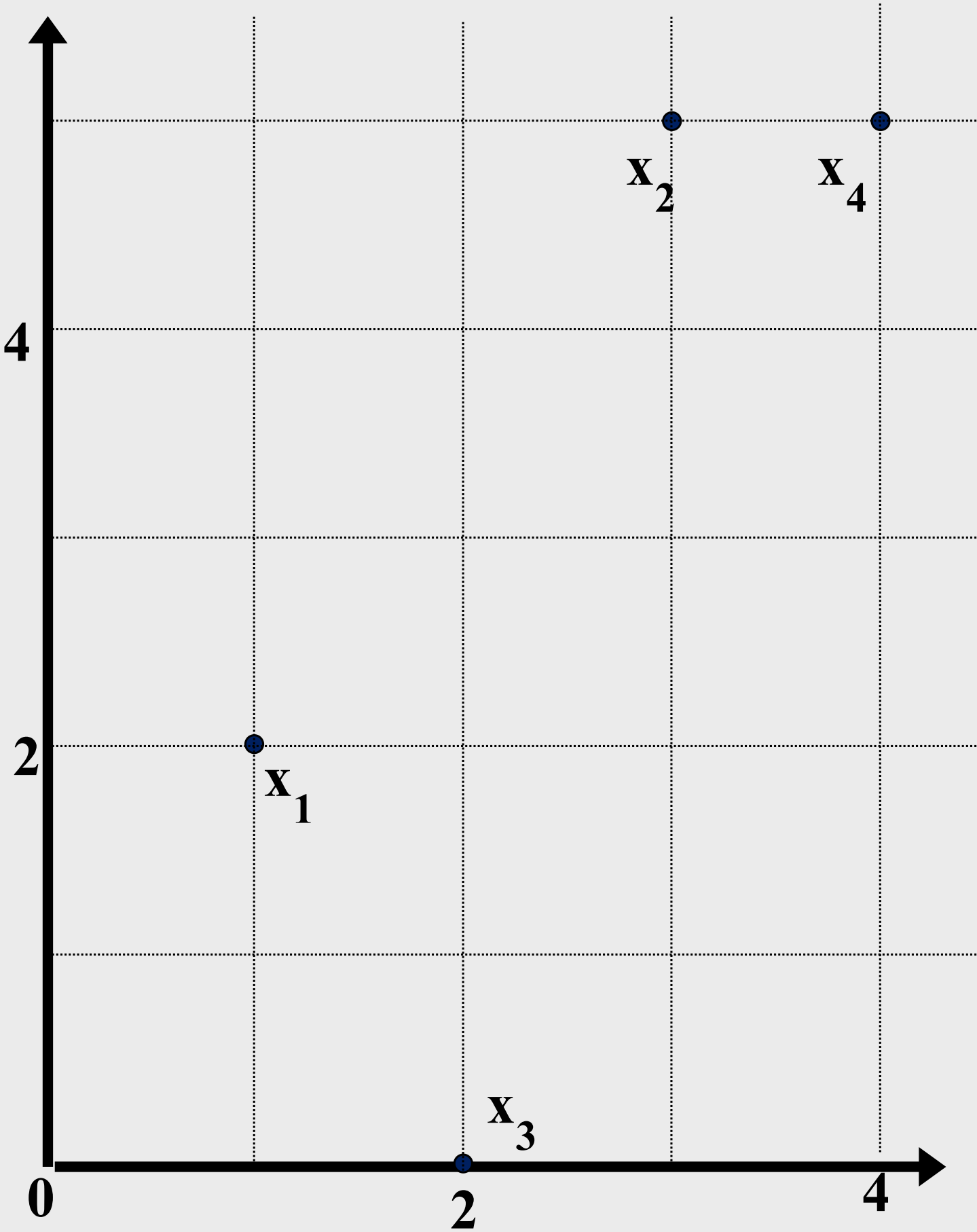
Dissimilarity matrix
(with **Euclidean**)

Dissimilarity matrix
(with **Manhattan**)

L	x_1	x_2	x_3	x_4
x_1	0			
x_2	5	0		
x_3	3	6	0	
x_4	6	1	7	0

L_∞	x_1	x_2	x_3	x_4
x_1	0			
x_2	3	0		
x_3	2	5	0	
x_4	3	1	5	0

Dissimilarity matrix
(with **supremum**)





PROXIMITY MEASURE FOR ORDINAL ATTRIBUTES

- An ordinal attribute can be discrete or continuous
- Order is important, e.g., rank
- Similar to numeric attribute that can be treated like interval-scaled:
 - Replace x_{if} by their rank $r_{if} \in \{1, \dots, M_f\}$
 - Map the range of each attribute onto $[0, 1]$ by replacing i -th object in the f -th attribute by

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

- Compute the dissimilarity using methods for interval-scaled attributes



PROXIMITY MEASURE FOR MIXED ATTRIBUTES TYPE

- A database may contain all attribute types
 - Nominal, symmetric binary, asymmetric binary, numeric, ordinal
- One may use a weighted formula to combine their effects

$$d(i, j) = \frac{\sum_{f=1}^p \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p \delta_{ij}^{(f)}}$$

- f is binary or nominal: $d_{ij}^{(f)} = 0$ if $x_{if} = x_{jf}$, or $d_{ij}^{(f)} = 1$ otherwise
- f is numeric: use the normalized distance
- f is ordinal:
 - Compute ranks r_{if} and $z_{if} = \frac{r_{if}-1}{M_f-1}$
 - Treat z_{if} as interval-scaled

k -MEANS AND k -MEDOID



PARTITIONING ALGORITHMS

- Partitioning method: partitioning a database D of n objects into a set of k clusters, such that the sum of squared distances is minimized (where c_i is the centroid or medoid of cluster C_i)

$$E(D, \mu) = \sum_{j=1}^k \sum_{i=1}^N d(x_i, \mu_j)^2$$

- The clustering find a partition of k clusters that optimizes the certain partitioning criterion. Methods:
 - **k -Means** (MacQueen' 67, Lloyd' 57/'82): each cluster is represented by center of cluster
 - **k -Medoids** or PAM (Kaufmann & Rousseeuw' 87): each cluster is represented by one of the objects in the cluster
 - Exhaustively enumerate all partitions



***k*-MEANS CLUSTERING**

- Creates an initial set of k partitions
- It then uses an iterative relocation technique that attempts to improve the partitioning by moving objects from one group to another.
- Each cluster is associated with a **centroid** (center point)
- Clustering: each point is assigned to the cluster with the closest centroid

```
1: Select  $K$  points as the initial centroids
2: repeat
3:     Form  $K$  clusters by assigning all points to the closest centroid
4:     Recompute the centroid of each cluster
5: until The centroids do not change
```

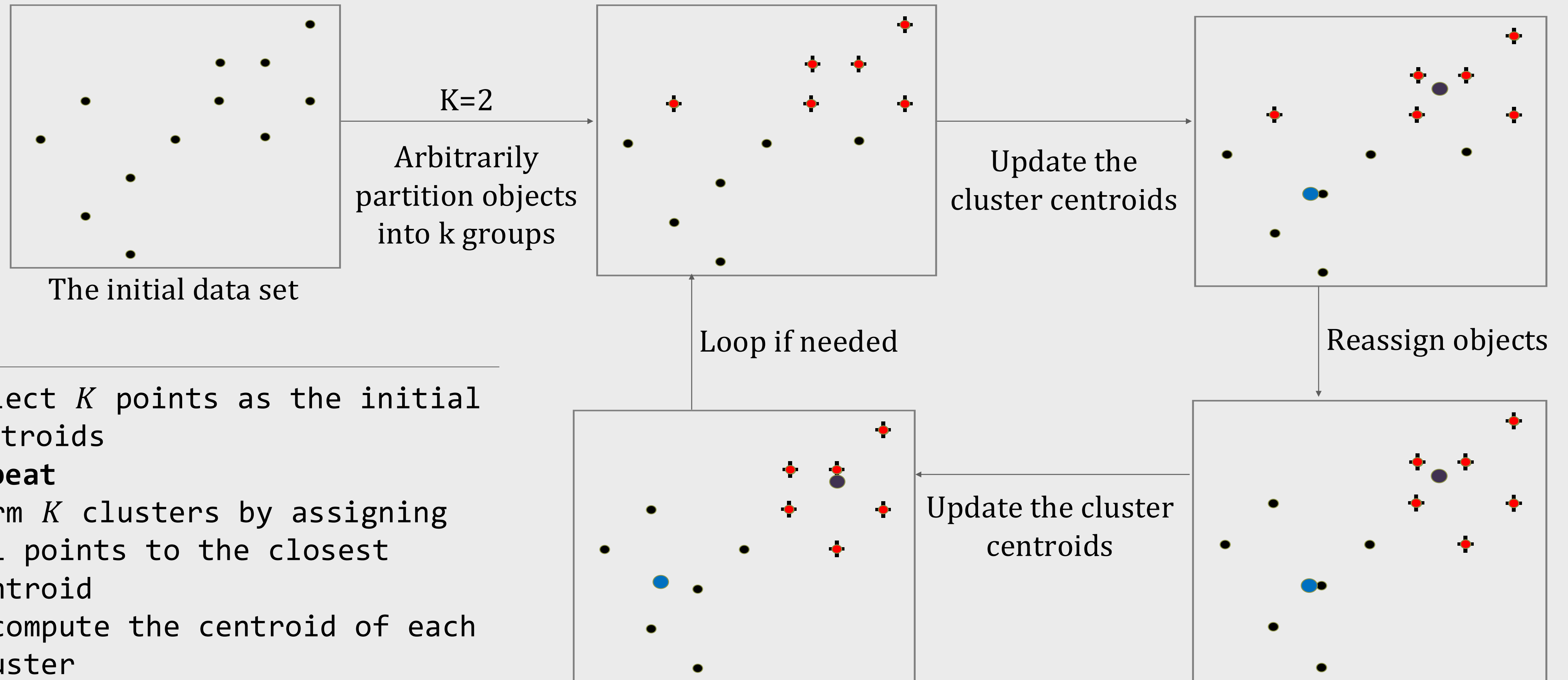


***k*-MEANS CLUSTERING (2)**

- Initial centroids are often chosen randomly
 - Clusters produced vary from one run to another
- The centroid is (typically) the mean of the points in the cluster
- ‘Closeness’ is measured by **Euclidean distance, cosine similarity, correlation**, etc.
- *k*-means will converge for common similarity measures mentioned above
- Most of the convergence happens in the first few iterations
 - Often the stopping condition is changed to ‘Until relatively few points change clusters’
- Complexity is $O(n * K * I * d)$
 - n = number of points, K = number of clusters,
 - I = number of iterations, d = number of attributes



k-MEANS CLUSTERING: EXAMPLES



- 1: Select K points as the initial centroids
- 2: **repeat**
- 3: Form K clusters by assigning all points to the closest centroid
- 4: Recompute the centroid of each cluster
- 5: **until** The centroids do not change



COMMENTS ON THE *k*-MEANS METHOD

- **Strength:** Efficient: $O(tkn)$, where n is # objects, k is # clusters, and t is # iterations. Normally, $k, t \ll n$.
 - Comparing: PAM: $O(k(n-k)^2)$, CLARA: $O(ks^2 + k(n-k))$
- **Comment:** Often terminates at a local optimal
- **Weakness:**
 - Applicable only to objects in a continuous n -dimensional space
 - Using the k -modes method for categorical data
 - In comparison, k -medoids can be applied to a wide range of data
 - Need to specify k , the number of clusters, in advance (there are ways to automatically determine the best k (see Hastie et al., 2009))
 - Sensitive to noisy data and outliers
 - Since an object with an extremely large value may substantially distort the distribution of the data
 - Not suitable to discover clusters with non-convex shapes



VARIATIONS OF THE *k*-MEANS METHOD

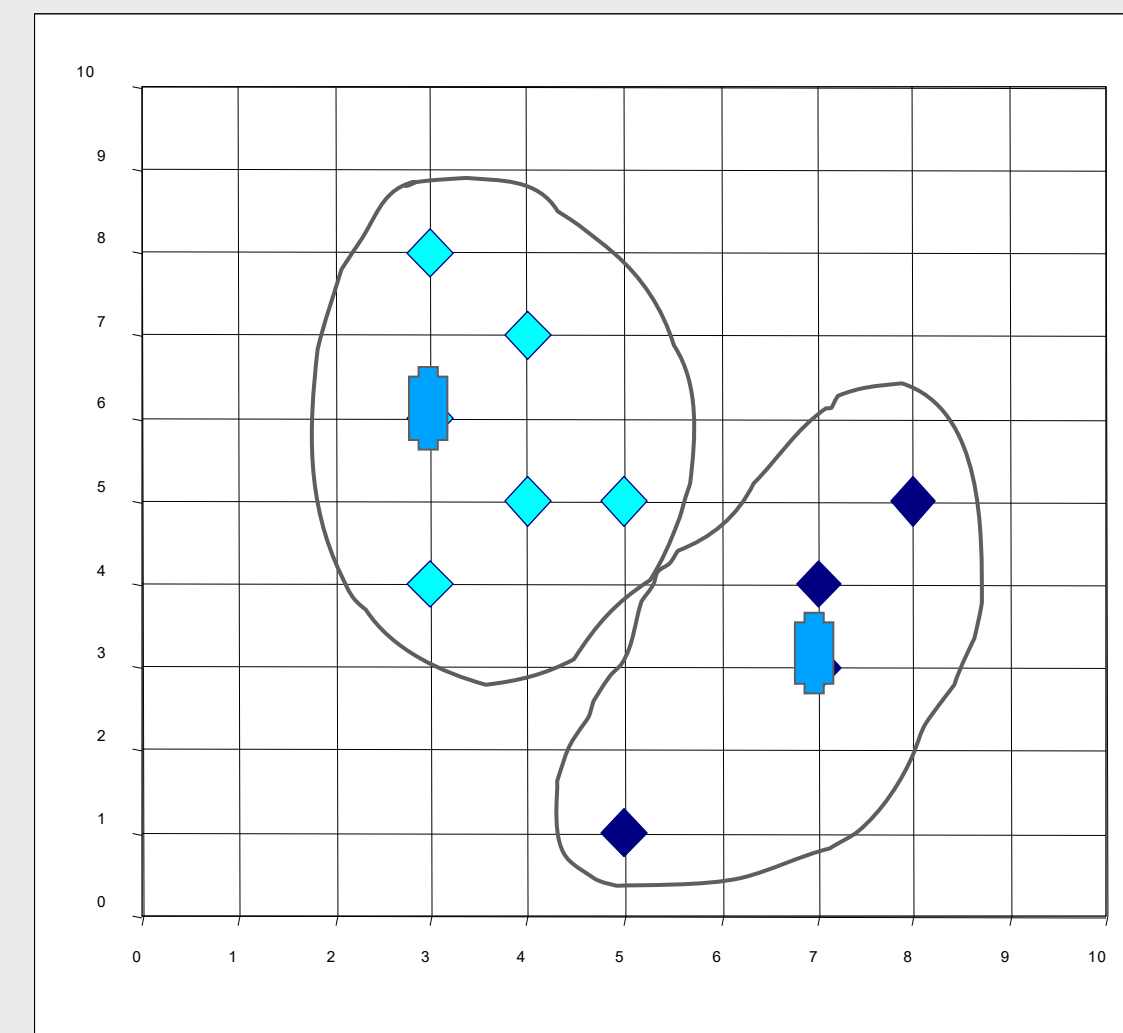
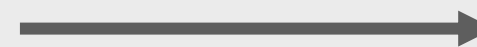
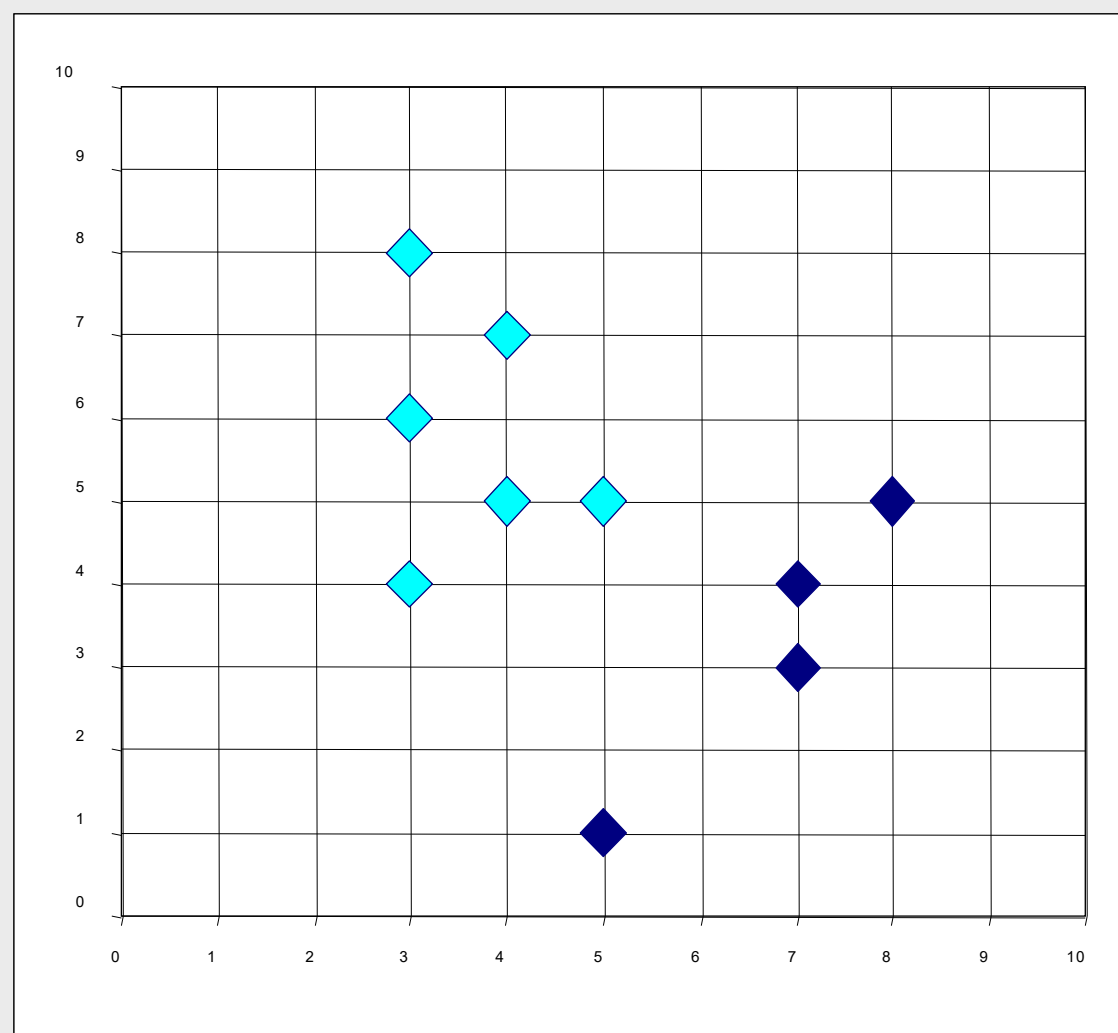
- Most of the variants of the *k*-means which differ in
 - Selection of the initial *k* means
 - Dissimilarity calculations
 - Strategies to calculate cluster means
- Handling categorical data: *k*-modes
 - Replacing means of clusters with modes
 - Using new dissimilarity measures to deal with categorical objects
 - Using a frequency-based method to update modes of clusters
 - A mixture of categorical and numerical data: *k*-prototype method

k -MEDOID



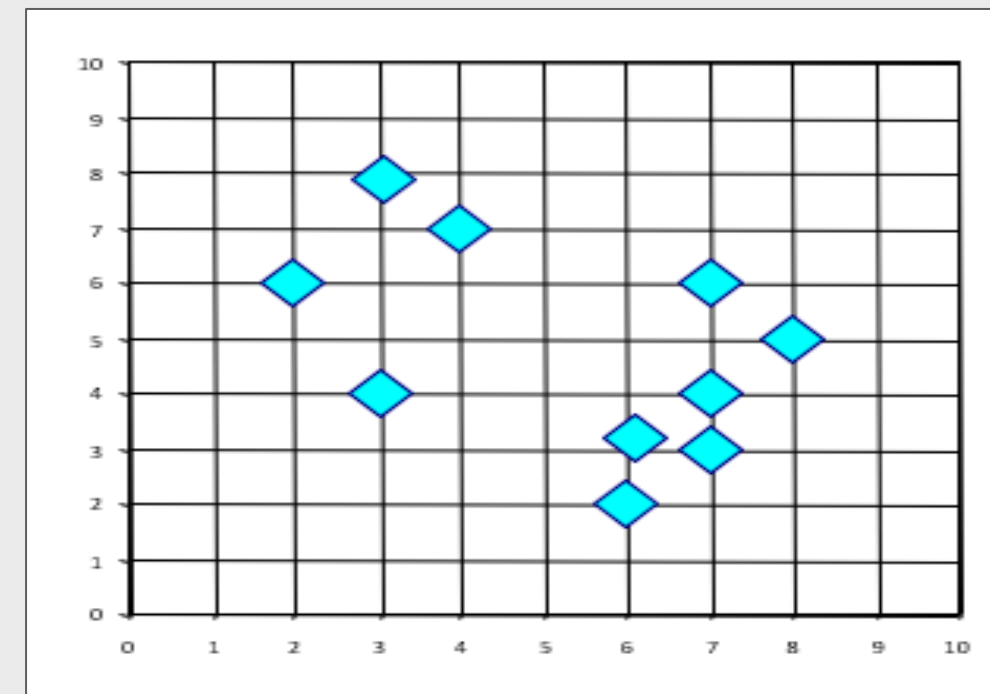
***k*-MEDOIDS**

- Instead of taking the mean value of the object in a cluster as a reference point, **medoids** can be used, which is the **most centrally located** object in a cluster

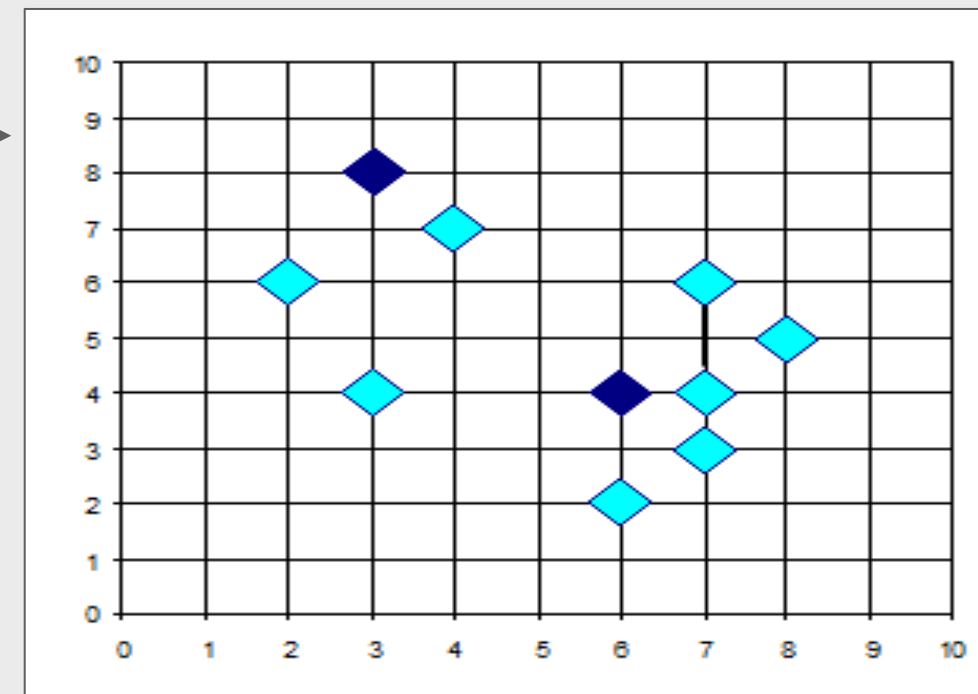




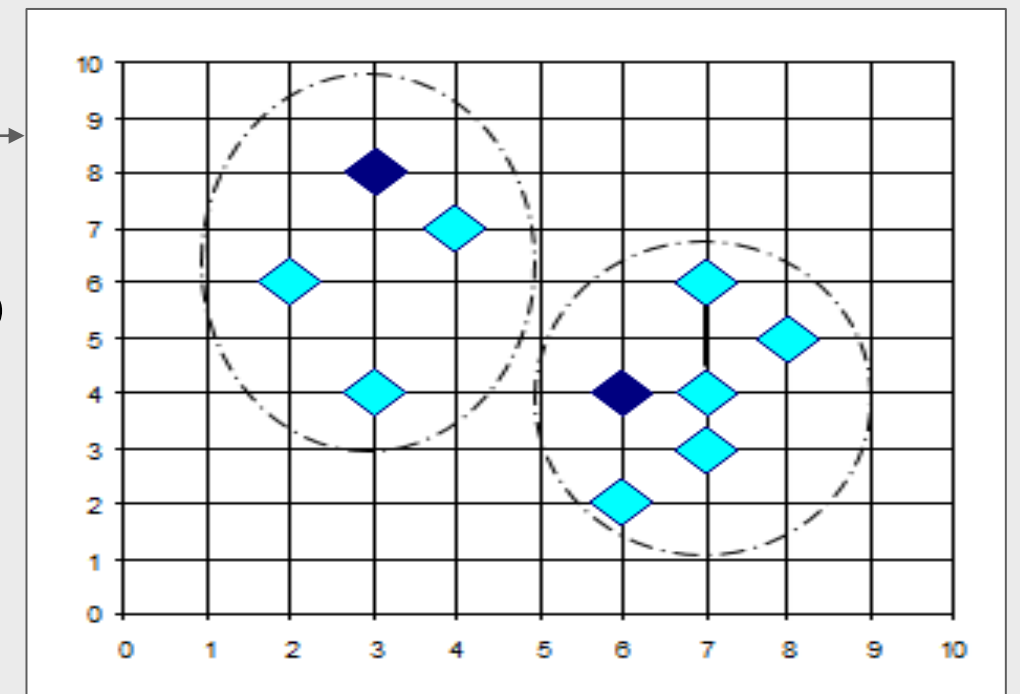
***k*-MEDOIDS CLUSTERING ALGORITHM**



Arbitrary choose k
object as initial
medoids

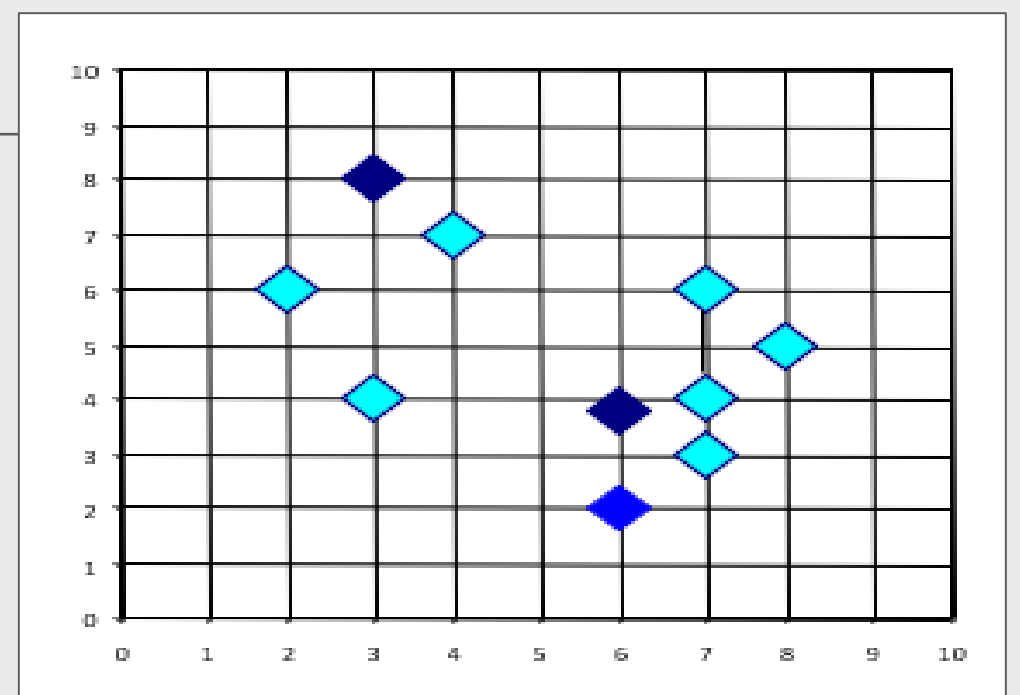


Assign each
remaining object to
nearest medoids

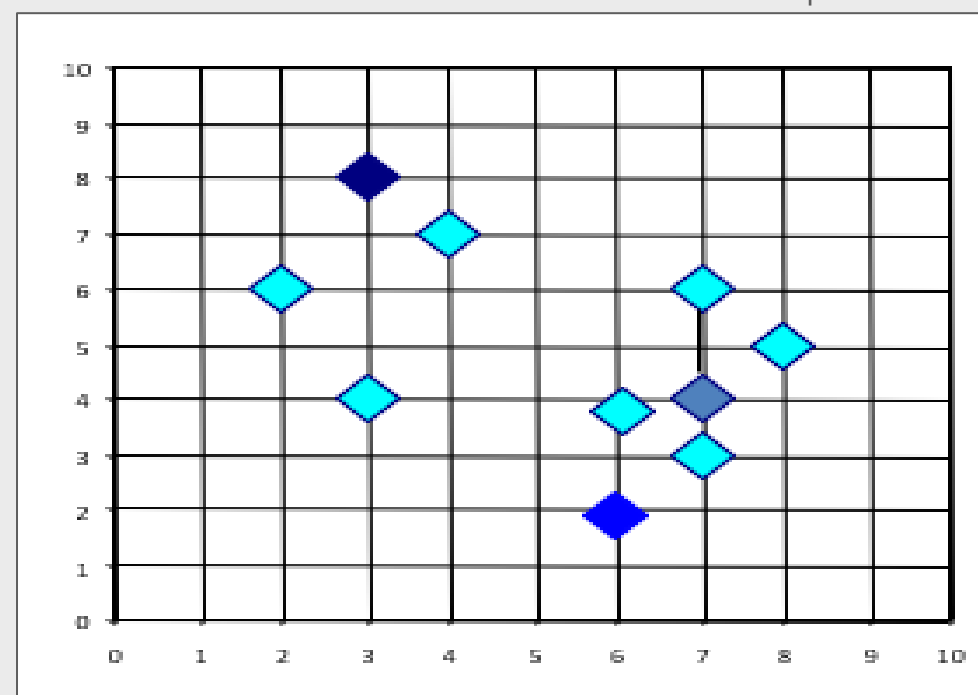


Total Cost = 20

Randomly select a non-
medoid object, O_{random}



Compute total cost
of swapping



Total Cost = 26

Swapping O
and O_{random}
If quality is
improved.

$K = 2$

Do loop

Until no change



***k*-MEDOIDS CLUSTERING ALGORITHM (2)**

- *K-Medoids* Clustering: Find representative objects (**medoids**) in clusters
 - *PAM* (Partitioning Around Medoids, Kaufmann & Rousseeuw 1987)
 - Starts from an initial set of medoids, and iteratively replaces one of the medoids by one of the non-medoids if it improves the total distance of the resulting clustering
 - *PAM* works effectively for small datasets, but does not scale well for large datasets (due to the computational complexity)
- Efficiency improvement on PAM
 - *CLARA* (Kaufmann & Rousseeuw, 1990): PAM on samples
 - *CLARANS* (Ng & Han, 1994): Randomized re-sampling

SUMMARY

- **Cluster analysis** groups objects based on their **similarity** and has wide applications
- Measure of similarity can be computed for **various types of data**
- Clustering algorithms can be **categorized** into partitioning methods, hierarchical methods, density-based methods, grid-based methods, and model-based methods
- **K-means** and **K-medoids** algorithms are popular partitioning-based clustering algorithms

EXERCISE

1. What is clustering in data mining? Explain with a simple real-life example.
2. Mention two common applications of clustering in real-world scenarios.
3. Suppose you are using K-Means clustering with $K = 3$. What will happen if the initial centroids are poorly chosen?

REFERENCES

Han, J., Kamber, M., & Pei, J. (2023). Data Mining: Concepts and Techniques (4rd ed.). San Francisco, CA, USA: Morgan Kaufmann Publishers Inc.